MILITARY STANDARD

MIL-STD-1320-176 (NAVY)

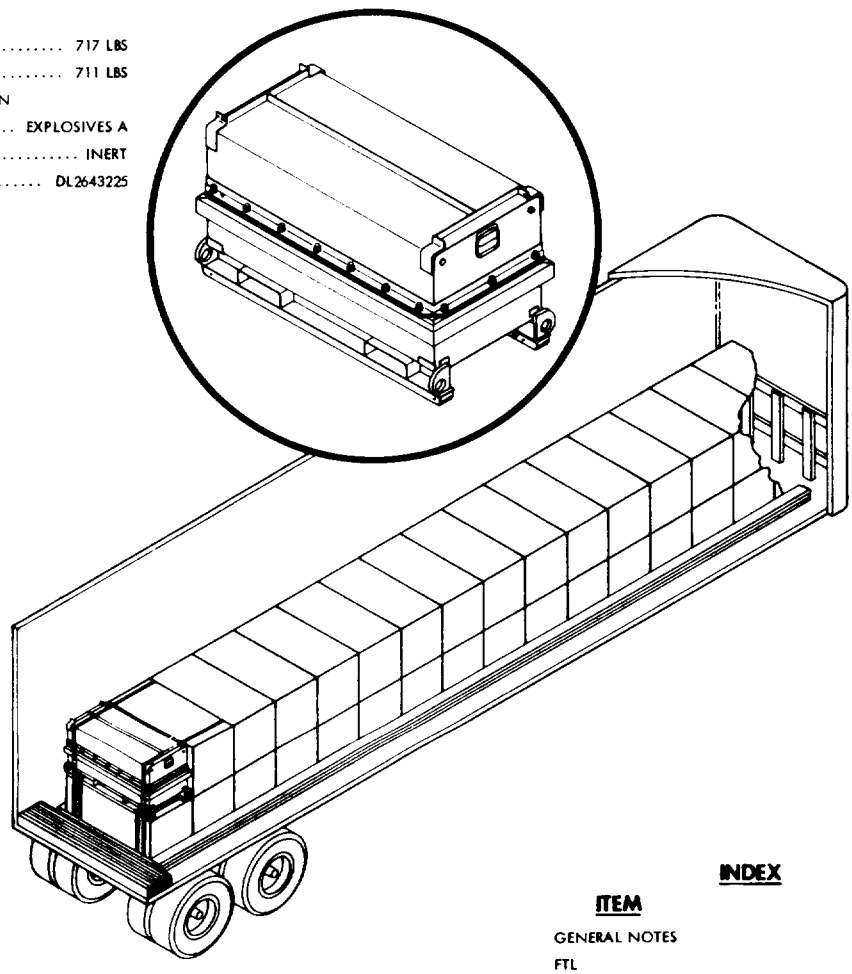
10 JANUARY 1977

TRUCKLOADING

WARHEAD, TACTICAL, WAU-3(V)/B (HARPOON) EXERCISE SECTION, AN/DTK-46 (HARPOON) IN CONTAINER MK 592 MOD 0

CONTAINER DATA

DIMENSIONS .... L 57 1/4, W 28 3/4, H 30 1/4
CUBE ..... 28.8 CU FT
GROSS WT
WARHEAD ..... 717 LBS
EXERCISE SECTION ..... 711 LBS
DOT HAZARD CLASSIFICATION
WARHEAD ..... EXPLOSIVES A
EXERCISE SECTION ..... INERT
DWG NO. DL2643225



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NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

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TECHNICAL DIRECTION AGENT (TDA) DATE 1/7/77

SEASYSCOM BY DIRECTION DATE 1/6/77

ORIGINATOR DATE 1/7/77

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE HARPOON TACTICAL WARHEAD WAU-3 (V)/B AND THE HARPOON EXERCISE SECTION AN/DKT-46 WHEN IN THE CONTAINER MK 592 MOD 0.
2. THESE PROCEDURES AND PRACTICES ARE INTENDED FOR 40, 42, 44 AND 45 FOOT VAN TRAILERS WITH NAILABLE OR NON-NAILABLE FLOORS. ALL TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. THE CONTAINERS WHEN STACKED TWO HIGH SHALL BE STRAPPED OR BOLTED TOGETHER INTO UNITS OF TWO AS SHOWN ON PAGE 5.
4. WHEN THE CONTAINERS ARE TWO LAYERS HIGH AT THE REAR OF THE TRAILER, THE LAST TWO STACKS SHALL BE BUNDLED TOGETHER AS SHOWN ON PAGE 3 USING TWO 1 1/4 X .035 STEEL STRAPS. THESE STRAPS PASS UNDER THE SKIDS OF THE BOTTOM CONTAINERS ON OVER THE TOPS OF THE UPPER CONTAINERS. STRAP MUST BE PREPOSITIONED IN THE TRAILER PRIOR TO LOADING THE CONTAINERS.
5. THE REAR BLOCKING SHIM, DETAIL A, IS POSITIONED BETWEEN THE CONTAINERS AND THE REAR BLOCKING ASSEMBLY WITH THE PIECE 2 BETWEEN THE CONTAINER'S LIFTING EYES. IF THE DISTANCE BETWEEN THE SHIM AND THE TRAILER DOORS IS
   A. LESS THAN 12 INCHES, USE REAR BLOCKING ASSEMBLY, DETAIL B
   B. GREATER THAN 12 INCHES, USE REAR BLOCKING ASSEMBLY, DETAIL G, PAGE 5
6. THE REAR BLOCKING ASSEMBLIES, DETAILS B AND G, SHALL FILL THE VOID BETWEEN THE REAR BLOCKING SHIM DETAIL A AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.

   WARNING

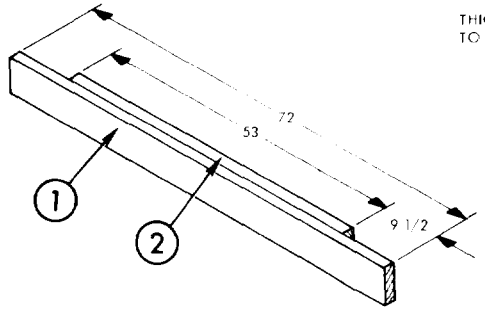
   DO NOT USE TRAILERS WITH ROLL UP DOORS WHEN BLOCKING BY THIS METHOD.
7. WHEN A TRAILER HAS NON-NAILABLE FLOORS AND SLEEPERS CANNOT BE NAILED TO THE FLOOR, SUBSTITUTE FILLER ASSEMBLY, DETAIL F, FOR THE SLEEPERS. THE FILLER ASSEMBLIES SHALL FILL THE VOID BETWEEN THE LADING AND THE TRAILER WALLS. ONE-HALF INCH SLACK IS PERMITTED.
8. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
9. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS. ATTACH "EXPLOSIVE A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER WHEN SHIPPING THE WARHEAD.
10. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT. STRAPPING, QQ-S-781, TYPE I CLASS A.
11. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40, 42, 44 & 45 FT TRAILERS (VAN)

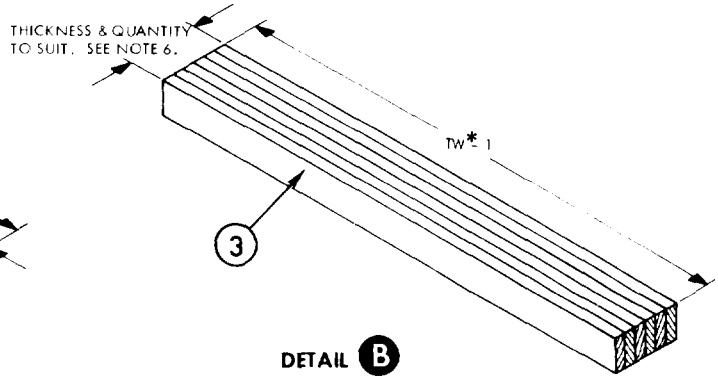
1. A FULL TRUCKLOAD FOR A 40 FT TRAILER IS 32 CONTAINERS. A FULL TRUCKLOAD FOR OTHER LENGTH TRAILERS IS:

TRAILER LENGTH	NO. OF CONTAINERS	TRUCKLOAD WT (APPROX.)	
		WARHEAD	EXERCISE SECTION
42 FT	34	24,378 LBS	24,174 LBS
44 FT	34	24,378 LBS	24,174 LBS
45 FT	36	25,812 LBS	25,596 LBS

2. BLOCK AND BRACE ALL LENGTH TRAILERS USING THE PRINCIPLES SHOWN FOR THE 40 FT TRAILER SHOWN ON PAGE 3 AND GENERAL NOTE 5.



DETAIL A
REAR BLOCKING SHIM
1 REQUIRED

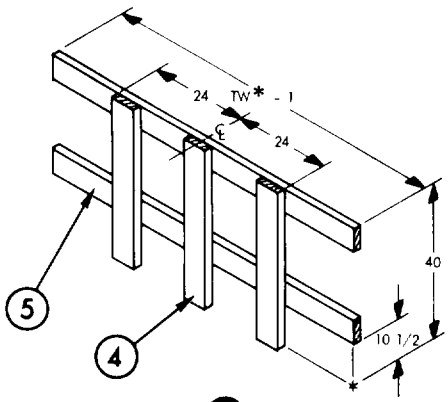


DETAIL B
REAR BLOCKING ASSEMBLY
1 REQUIRED

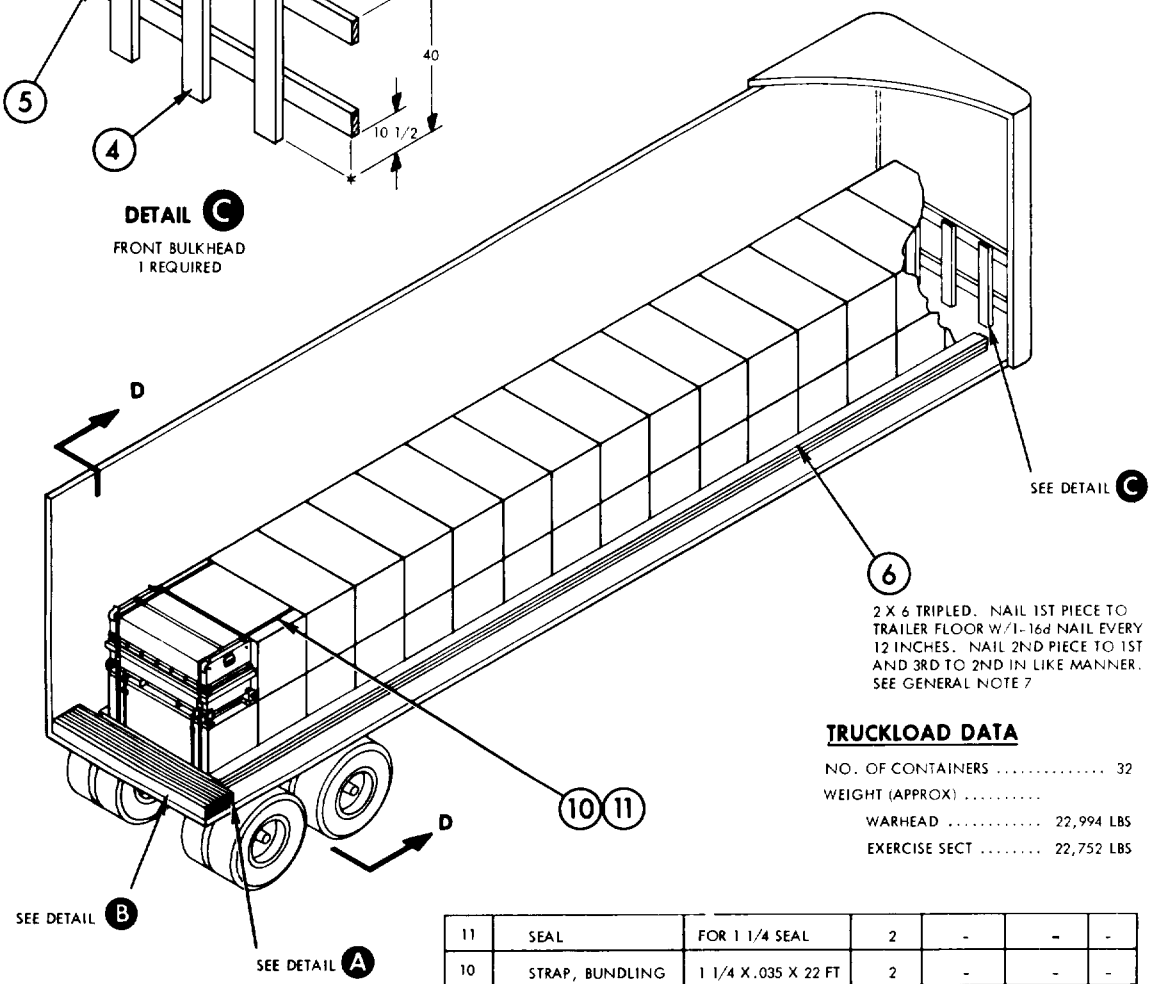
* TW = TRAILER WIDTH

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* TW TRAILER WIDTH (AT FORWARD WALL)



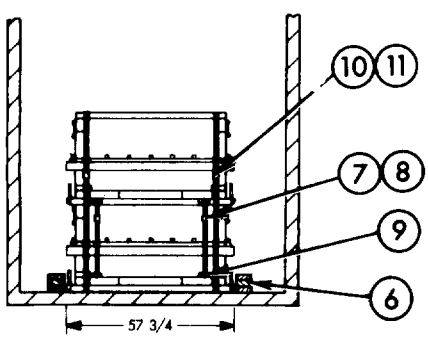
DETAIL C
FRONT BULKHEAD
1 REQUIRED



2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 12 INCHES. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER. SEE GENERAL NOTE 7

TRUCKLOAD DATA

NO. OF CONTAINERS 32
WEIGHT (APPROX) .....
WARHEAD ..... 22,994 LBS
EXERCISE SECT ..... 22,752 LBS



(CENTER LOAD IN TRAILER)
SECTION D-D

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS: NUMBER	NAILS: SIZE
11	SEAL	FOR 1 1/4 SEAL	2	-	-	-
10	STRAP, BUNDLING	1 1/4 X .035 X 22 FT	2	-	-	-
9	EDGE PROTECTOR	FOR 3/4 STRAP	128	-	-	-
8	SEAL	FOR 3/4 SEAL	32	-	-	-
7	STRAP	3/4 X .035 X 12 FT	32	-	-	-
6	SLEEPER	2 X 6 X RANDOM LENGTHS	231 FT	SEE FIELD NOTE		
5	HORIZONTAL	2 X 6 X TW* - 1	2	SEE 4	-	-
4	VERTICAL	2 X 6 X 40	3	5	4 PER JOINT	10d
3	HORIZONTAL	THICKNESS TO SUIT 6 WIDE	TO SUIT	LAMINATE	6 PER PIECE	TO SUIT
2	SHIM	2 X 6 X 53	1	1	5	10d
1	SHIM	2 X 6 X 72	1	DETAIL B OR G	2 PER JOINT	10d

LIST OF MATERIALS & NAILING DATA

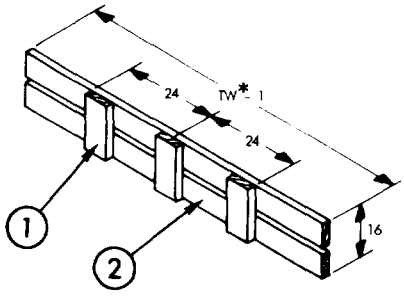
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LTL ALL LENGTH TRAILERS (VAN)

1. WHEN REQUIRED TO SHIP LESS THAN TRUCKLOAD, DECREASE THE NUMBER OF CONTAINERS IN THE SECOND LAYER STARTING AT THE REAR OF THE LOAD AND WORKING FORWARD. BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGE 3.
2. WHEN THE SECOND LAYER HAS BEEN REMOVED, DECREASE THE NUMBER OF CONTAINERS IN THE BOTTOM LAYER STARTING AT THE REAR AND WORKING FORWARD. BLOCK AND BRACE AS SHOWN FOR THE LTL SHOWN BELOW.

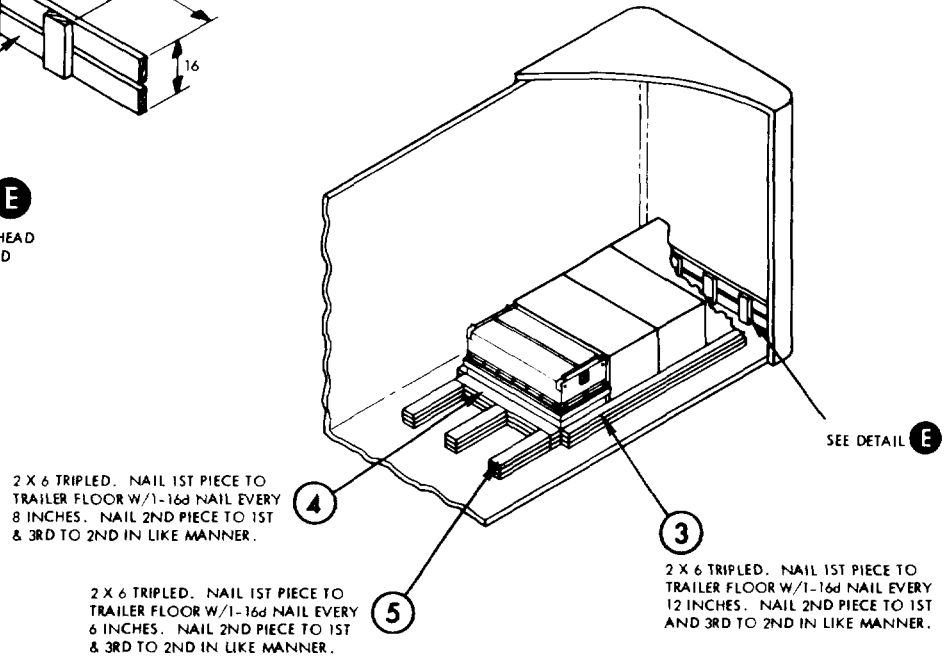
WARNING

DO NOT USE TRAILERS WITH NON-NAILABLE FLOOR WHEN BLOCKING BY THIS METHOD. TRAILERS MUST HAVE NAILABLE FLOORS.



DETAIL E

FRONT BULKHEAD
1 REQUIRED



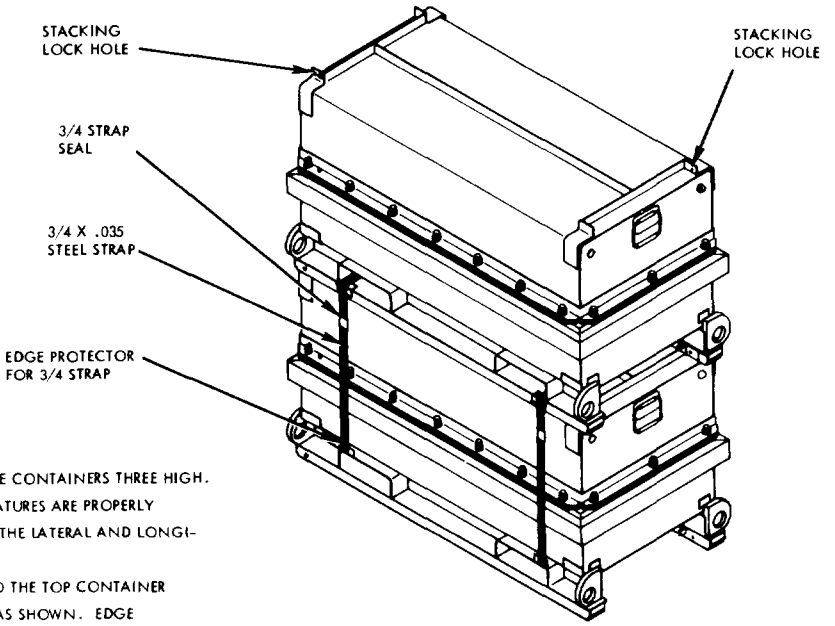
*TW TRAILER WIDTH (AT FORWARD WALL)

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS: NUMBER	NAILS: SIZE
LIST OF MATERIALS & NAILING DATA						
1	VERTICAL	2 X 6 X 16	3	2	4 PER JOINT	10d
2	HORIZONTAL	2 X 6 X TW* - 1	2	SEE 1	-	-
3	SLEEPER	2 X 6 X RANDOM LENGTHS	58 FT	SEE FIELD NOTE		
4	CROSSMEMBER	2 X 6 X 57	3	SEE FIELD NOTE		
5	BACKUP CLEAT	2 X 6 X 30	9	SEE FIELD NOTE		

TYPICAL LTL (VAN)

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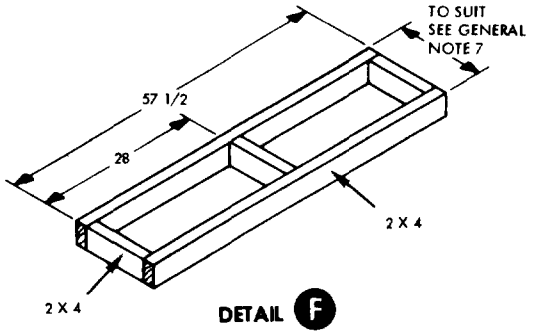
UNITIZING PROCEDURE



1. USING A FORKLIFT TRUCK STACK THE CONTAINERS THREE HIGH.
2. BE CERTAIN THAT THE STACKING FEATURES ARE PROPERLY ALIGNED TO PREVENT SHIFTING IN THE LATERAL AND LONGITUDINAL DIRECTION.
3. SECURE THE BOTTOM CONTAINER TO THE TOP CONTAINER WITH TWO 3/4 X .035 STEEL STRAPS AS SHOWN. EDGE PROTECTORS SHALL BE USED UNDER THE STRAPS (4 PLACES).
4. TENSION STRAPS AND SECURE WITH ONE DOUBLE NOTCHED SEAL OR TWO DOUBLE CRIMPED SEALS.
5. IF DESIRED, THE STACK OF CONTAINERS MAY BE SECURED TOGETHER USING 3/8 X 4 BOLTS WITH WASHERS AND ELASTIC STOP NUTS. TWO BOLTS ARE USED TO SECURE ONE LAYER TO THE OTHER, THE BOLTS PASSING THROUGH THE STACKING LOCK HOLES IN DIAGONALLY OPPOSITE CORNERS OF THE CONTAINERS. CONTAINERS BOLTED TOGETHER ARE NOT REQUIRED TO BE STRAPPED.

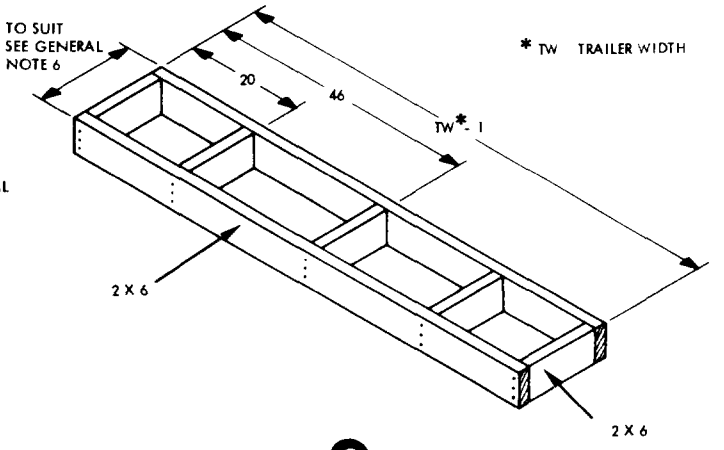
WARNING

THE STACKED CONTAINERS SHALL BE BROKEN DOWN IMMEDIATELY UPON UNLOADING THE TRAILER.



DETAIL F

FILLER ASSEMBLY
(FOR TRAILERS WITH NON-NAILABLE FLOORS)



DETAIL G

REAR BLOCKING ASSEMBLY

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